



Azure Migration Plan

Full migration of the TAQA Knowledge Hub onto TAQA Azure, including the Microsoft 365 services used at each phase.

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INTERNAL USE

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1. What is being migrated

The Knowledge Hub is a folder of files. That is the whole thing.

33 files, about 2.3 MB. No database, no server, no backend. No build step, no compiling, no package manager. The files sitting in the repository are exactly the files served to the browser.

This is the most important fact for planning, because it means there is nothing to migrate except static files. No data to export, no server to rebuild, no dependencies to resolve, no downtime to schedule.

Type	Files	What they are
Pages	11 <code>.html</code>	Home, segment view, document viewer, library, search, glossary, upload, support, control panel, analytics, offline page
Scripts	5 <code>.js</code>	Shared UI, two data files, the QR library and the service worker
Fonts	13 files	TAQA corporate typeface plus self-hosted web fonts
Images	5 plus icons	Hero images and app icons
Config	<code>manifest.json</code> , <code>staticwebapp.config.json</code>	App install settings and Azure routing rules

The current state of the application, and what it does and does not do, is covered in the companion document "What the platform does".

2. Microsoft 365 and Azure services used

Every service below is already owned by TAQA. **The migration requires no new licences and no new products.**

Phase 1: identity

Service	Role in the platform	Licence position
Microsoft Entra ID	Provides sign-in. Replaces having no login at all. Supplies the user's name, email and group membership to the platform	Included in existing TAQA M365
Entra ID App Registration	The registered identity of the platform, so Entra will issue it tokens	No cost
Entra security groups	Map the 12 segments to sets of users, which is how per-segment access is enforced later	Included
Azure Static Web Apps	Hosts the site and handles the sign-in redirect natively	Free tier is \$0. Standard tier is recommended for the SLA and custom domains

Phase 2: document storage

Service	Role in the platform	Licence position
SharePoint Online	Holds the actual document files. Becomes the source of truth, replacing the placeholder in the viewer	Included in existing TAQA M365
Microsoft Graph API	The interface the platform uses to read document metadata and files from SharePoint	No cost
SharePoint metadata columns	Segment, document type, revision, owner and review date, so the platform's filters read live data rather than a static file	Included
Microsoft Search	Optional. Can back the cross-segment search with a real index instead of the current fixed list	Included

Phase 3: workflow

Service	Role in the platform	Licence position
Power Automate	Runs the document upload and approval flow, and the support ticket flow. Turns the two forms that currently do nothing into working processes	Included in M365. Premium connectors would need review, but this design does not require them
Microsoft Lists or SharePoint lists	Holds support tickets and the approval queue	Included
Exchange Online	Sends the notification emails the flows produce	Included

Phase 4: analytics, notifications and protection

Service	Role in the platform	Licence position
Application Insights	Real usage analytics, replacing the per-browser counters	Consumption based, minimal at this volume
Microsoft Purview	Sensitivity labels, data loss prevention and retention on the documents	Included in existing TAQA M365
Microsoft Defender for Cloud Apps or Defender for Storage	Malware scanning on uploaded files. This is the control that makes the upload path safe	Subject to TAQA's existing Defender coverage
Azure Key Vault	Optional. Holds the Entra client secret if D&T prefers it over Static Web Apps configuration	Minimal

Phase 5: operations

Service	Role in the platform	Licence position
Azure Monitor	Uptime and availability monitoring	Consumption based
Microsoft Sentinel	Security logging and alerting, if TAQA routes application logs there	Per TAQA's existing arrangement
Azure DevOps or GitHub Enterprise	Source control and the deployment pipeline under TAQA governance	Per TAQA's existing arrangement

What is not needed

No virtual machines. No App Service. No SQL or Cosmos database. No storage account. No Application Gateway. No API Management. No container registry.

If a design or a quote comes back including any of those, the scope has been misread. This is a static site reading from SharePoint.

3. Target architecture

The finished state, in one paragraph. A TAQA employee opens the Hub's address. Azure Static Web Apps sees no valid session and redirects them to Microsoft Entra ID. They sign in with their normal TAQA account, with whatever MFA and conditional access policies TAQA already enforces. Entra returns them to the Hub with a token. The Hub reads their name, email and group membership, and uses the group membership to decide which segments they may see. When they open a document, the Hub calls Microsoft Graph to fetch it from SharePoint, and SharePoint's own permissions apply as a second check. When they upload a document, it lands in SharePoint, Defender scans it, and Power Automate routes it to the segment owner for approval.

At no point does the platform hold credentials, store documents itself, or call anything outside the TAQA tenant.

Trust boundaries

Boundary	What crosses it	Control
Browser to Azure Static Web Apps	Page requests, authenticated session cookie	HTTPS, HSTS, Entra session
Static Web Apps to Entra ID	Authentication redirect and token exchange	Managed natively by the platform. The client secret is held in configuration, never in code
Browser to Microsoft Graph	Document metadata and file requests, carrying the user's own token	Delegated permissions, so a user can only reach what they could reach in SharePoint directly
Power Automate to SharePoint	Approval actions and file moves	Service account or managed identity, per D&T standard

Delegated permissions matter here. The platform never holds elevated rights of its own. It acts as the signed-in user, so SharePoint's permission model remains the single source of truth for who can see what.

4. Prerequisites

Required to host it

#	What	Notes
1	Resource group in the TAQA tenant	With a cost centre assigned
2	Azure Static Web App resource	Standard tier recommended, needed for the SLA and custom domains. Free tier works for a pilot
3	Deployment token stored as a repository secret	Azure issues this when the resource is created
4	Repository under the TAQA organisation	GitHub Enterprise or Azure DevOps, D&T's choice
5	A URL	Azure-generated address, or a TAQA custom domain

Required for sign-in

#	What	Notes
6	Entra ID App Registration	The main blocker. Nothing about login can start without it
7	Client ID and Tenant ID	Issued by the App Registration
8	Client secret in Static Web App configuration	Never in code. Key Vault is an alternative
9	Redirect URI registered	<code>https://<final-url>/.auth/login/aad/callback</code>
10	Graph permission <code>User.Read</code> with admin consent	The minimum for sign-in

Required for documents

#	What	Notes
11	SharePoint site with a library structure for the 12 segments	Metadata columns agreed first
12	Graph permissions <code>Sites.Read.All</code> and <code>Files.Read.All</code> with admin consent	Read only. Write scopes are added at Phase 3, not before
13	A permission model per segment	Decided with the segment owners

5. Migration steps

Six steps. Steps 1 to 4 complete the move. Steps 5 and 6 add sign-in.

Step 1: take the code

The repository transfers into the TAQA organisation, or D&T forks it, whichever matches policy. Full commit history comes across intact, which preserves the audit trail from the Secure SDLC review.

D&T receives the repository, this plan, the functional description, the SDLC assessment and the cybersecurity review pack.

Step 2: create the Azure resource

Create the Static Web App in the TAQA resource group and link it to the repository. Azure generates the deployment workflow automatically.

One setting causes most first-time failures, because there is no build step:

```
app_location:    "/"
api_location:    ""
output_location: ""    <-- must be EMPTY
```

If `output_location` is set to `dist` or `build`, the deployment fails looking for a folder that does not exist. There is no build output. The source is the site.

Step 3: deploy and verify

Push to `main` and the site deploys in about two minutes.

Check six things:

1. The home page loads with the TAQA typeface, not a fallback
2. Navigation between pages works
3. Search returns results
4. The Arabic toggle switches the layout right to left
5. With the network switched off, the site still loads
6. The browser console shows no errors and no requests to any outside host

`staticwebapp.config.json` is already in the repository and handles routing, font MIME types, caching and the security headers. No configuration is needed.

Step 4: lock the code down

Enable branch protection on `main`: no direct pushes, pull request plus one approval required. Nominate the D&T technical reviewer as approver.

This closes the code review control from the Secure SDLC assessment and is the point at which the platform comes formally under D&T governance.

At this stage the migration is complete. The site runs on TAQA Azure, under TAQA source control, with a review gate.

Step 5: add Entra ID sign-in

Create the App Registration, store the client secret in the Static Web App configuration, then add this to

`staticwebapp.config.json`:

```

"auth": {
  "identityProviders": {
    "azureActiveDirectory": {
      "registration": {
        "openIdIssuer": "https://login.microsoftonline.com/<TENANT_ID>/v2.0",
        "clientIdSettingName": "AAD_CLIENT_ID",
        "clientSecretSettingName": "AAD_CLIENT_SECRET"
      }
    }
  },
  "routes": [
    { "route": "/*", "allowedRoles": ["authenticated"] }
  ]
}

```

Sequence matters. Add this after the App Registration exists. Applying it first locks everyone out of a site that has no way to let them back in.

Azure Static Web Apps handles the sign-in flow natively, so no authentication code needs writing. The signed-in user's details become available to the pages at `/.auth/me`. The platform already contains the placeholder that reads this, so wiring it up is small.

Step 6: decide who gets in

Two options, to be decided with the business.

Open to all TAQA staff means anyone with a TAQA account can sign in and see everything. It is the simplest to run and the fastest to launch.

Restricted by segment means Entra security groups control which segments each person sees. It needs a decision per segment from the owners, and ongoing group administration as people move roles.

My recommendation is open to all staff at launch. The content is internal reference material, and per-segment restriction can be added afterwards once real usage shows whether it is needed. Starting restricted means 12 permission decisions before anyone can use the platform.

6. Phases after the migration

The migration itself ends at step 6. These are the phases that turn the platform into a full document system.

Phase	What it delivers	Main services	Indicative effort
1. Identity	TAQA sign-in on a TAQA-hosted site	Entra ID, Static Web Apps	2 to 3 days after approvals
2. Documents	Real files in the viewer, live metadata driving the lists and search	SharePoint, Graph	8 to 13 days
3. Workflow	Upload and support forms actually work, with approval routing	Power Automate, Lists, Exchange	8 to 13 days
4. Analytics and protection	Real usage figures, shared bookmarks, sensitivity labels, malware scanning	Application Insights, Purview, Defender	6 to 10 days
5. Operations	Monitoring, support model, penetration test, rollout	Azure Monitor, Sentinel	Per D&T

Effort figures are our estimate to anchor discussion. D&T's own sizing governs.

7. Security and compliance during migration

Item	Position
Data residency	The Azure region must satisfy TAQA's requirements. Worth confirming against NCA Cloud Cybersecurity Controls
Secrets	The Entra client secret lives in Static Web Apps configuration or Key Vault. Never in the repository. No secrets exist in the code today, verified across every commit
Least privilege	Graph permissions start read only. Write scopes are added at Phase 3 and not before
Public repository	The repository is public today because GitHub requires it for the free preview link. It should be made private once TAQA hosting is live
Offline caching	The platform caches pages on the device for offline reading. Before real documents are loaded, a decision is needed on whether caching operational documents onto personal devices is acceptable
Penetration testing	Should follow Phase 1 and Phase 2. Testing the current static site would examine a file server rather than the system that will go live
Existing findings	Four issues were found and closed during the security review. Details are in the Secure SDLC assessment and the cybersecurity review pack

8. Current status: the platform is offline

The Hub previously ran on a free-trial Azure subscription inside the TAQA directory (taqa.com.sa). That trial credit has expired, the subscription now shows as Disabled, and the Azure URL returns 404.

A GitHub Pages preview remains available in the meantime.

This is the practical reason the migration matters now rather than later. The resource itself runs on the Static Web Apps Free tier and carries no licence or hosting cost. What is needed is a TAQA-owned subscription to host it under.

9. What we need from D&T

1. Confirm GitHub Enterprise or Azure DevOps for the repository
 2. Confirm Standard or Free tier for the Static Web App
 3. Confirm the URL: Azure default or a TAQA custom domain
 4. Name the technical reviewer, so branch protection becomes meaningful
 5. Name the IT coordinator to co-own the integration phases
 6. Start the App Registration request. It has the longest lead time and blocks everything else
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10. Timeline

Stage	Effort	Waiting on
Steps 1 to 4, running on TAQA Azure	1 to 2 days	Resource group and repository decision
Steps 5 and 6, Entra ID sign-in live	2 to 3 days	App Registration approval
Phase 2 onward	See section 6	Phase 1 completion and business decisions

The engineering work is small. The waiting is on approvals.

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